

MERCED, CA, USA

WMO: 724815

Lat: 37.285N

Lon: 120.513W

Elev: 152

StdP: 14.62

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23257

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.7	32.1	21.5	16.2	48.7	25.4	19.6	46.5	24.7	53.8	20.6	52.2	2.4	100	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.2	101.5	69.5	99.0	68.7	96.7	67.6	72.2	96.0	70.6	94.1	69.1	92.2	9.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.5	81.7	61.5	81.9	79.9	59.9	77.5	79.2	35.9	96.1	34.4	94.0	33.2	92.3	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 19.5	16.8	14.4		26.0	106.2	2.4	2.0	24.3	107.6	22.9	108.8	21.5	109.9	19.8	111.3	
(5)			DB	24.6	75.4	2.8	4.1	22.6	78.4	21.0	80.8	19.4	83.1	17.4	86.1	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	46.4	50.2	55.2	59.8	67.5	74.8	79.5	77.7	73.4	64.1	53.5	46.5	(6)
	DBStd	12.86	4.84	4.68	5.50	6.35	6.22	6.00	4.95	4.29	5.87	5.60	5.54	5.18	(7)
	HDD50	365	131	51	16	4	0	0	0	0	0	0	28	135	(8)
	HDD65	2533	575	413	305	181	43	4	0	0	7	84	346	575	(9)
	CDD50	4907	21	57	177	297	544	743	914	858	702	436	133	25	(10)
	CDD65	1601	0	0	1	24	122	297	449	393	259	55	1	0	(11)
	CDH74	22521	0	0	95	507	1920	4216	6227	5229	3421	873	33	0	(12)
	CDH80	12184	0	0	12	164	841	2321	3739	3016	1806	282	3	0	(13)
(14) Wind	WSAvg	6.1	4.6	5.5	5.9	7.0	7.8	8.0	7.5	7.0	5.9	5.2	4.3	4.9	(14)
(15) Precipitation	PrecAvg	12.4	2.4	2.2	2.0	1.0	0.4	0.1	0.0	0.0	0.2	0.7	1.6	1.9	(15)
	PrecMax	21.7	7.1	6.1	5.5	4.6	2.1	0.6	0.4	0.5	1.3	3.8	4.9	4.5	(16)
	PrecMin	6.6	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
	PrecStd	3.9	1.8	1.6	1.5	0.9	0.6	0.2	0.1	0.1	0.3	0.9	1.3	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	71.5	81.5	91.1	97.5	103.9	105.3	103.3	100.3	92.9	78.2	65.0	(19)
		MCWB	53.2	57.7	61.3	64.4	67.2	70.6	71.1	69.5	67.7	65.1	60.2	55.0	(20)
	2%	DB	61.9	67.5	76.4	84.4	93.3	99.4	101.9	99.7	97.7	86.8	73.1	61.3	(21)
		MCWB	52.8	56.3	59.7	61.9	65.4	69.4	69.7	68.8	66.7	63.5	58.1	53.0	(22)
	5%	DB	58.7	64.0	72.6	80.1	89.0	96.2	99.2	97.2	94.3	82.3	69.3	58.5	(23)
		MCWB	52.0	54.5	58.4	60.5	63.5	67.4	69.4	67.8	65.9	61.4	56.9	52.1	(24)
	10%	DB	55.5	61.1	68.7	75.1	84.2	92.3	96.3	94.2	90.7	79.0	65.2	55.9	(25)
		MCWB	50.5	53.0	56.8	58.4	61.8	66.0	68.1	67.0	65.1	59.9	55.0	50.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	60.3	64.5	67.1	70.3	73.3	74.7	73.2	72.4	67.8	62.4	58.8	(27)
		MCDB	61.3	67.5	76.1	83.0	93.3	98.2	97.3	96.1	91.1	84.3	73.8	62.9	(28)
	2%	WB	55.0	58.0	61.6	63.8	67.1	71.0	72.4	71.0	69.3	65.4	59.9	55.7	(29)
		MCDB	58.7	64.7	73.6	81.2	89.8	95.4	97.3	94.7	90.5	82.2	70.4	58.9	(30)
	5%	WB	53.1	55.8	59.5	61.5	64.7	69.1	70.6	69.3	67.6	63.4	57.9	53.7	(31)
		MCDB	56.2	62.4	70.2	77.8	85.7	92.0	95.0	92.7	90.2	80.1	66.5	56.9	(32)
	10%	WB	51.3	54.0	57.4	59.5	62.7	67.4	69.1	68.0	66.0	61.4	56.1	51.6	(33)
		MCDB	55.2	59.8	66.9	73.2	82.2	89.7	92.8	91.0	88.1	75.6	63.6	54.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	20.5	23.5	25.8	28.8	31.4	32.2	32.3	31.5	27.6	22.4	17.8	(35)
		MCDBR	22.8	25.6	30.1	33.7	35.3	35.0	34.3	35.1	36.0	33.3	28.6	21.5	(36)
	5% WB	MCWBR	14.9	15.4	15.4	14.5	13.6	12.1	11.5	11.7	12.1	14.2	15.2	13.8	(37)
		MCWBR	16.6	22.3	27.6	31.2	33.2	33.3	32.0	33.1	33.1	29.2	24.2	14.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.347	0.352	0.359	0.361	0.361	0.352	0.366	0.355	0.356	0.367	0.355	0.335	(40)
		taud	2.384	2.393	2.369	2.381	2.402	2.436	2.409	2.417	2.425	2.378	2.373	2.445	(41)
	Ebn at Noon		263	278	286	292	292	293	288	290	284	268	257	260	(42)
		Edh at Noon	29	32	36	37	37	36	37	36	34	32	29	25	(43)
(44) All-Sky Solar Radiation	RadAvg		744	1093	1564	2003	2397	2663	2591	2351	1935	1404	935	683	(44)
	RadStd		119	121	140	160	121	84	80	66	61	80	56	99	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.72	N/A	-2.07	N/A	N/A	N/A	N/A	-205	+248	+63

Nomenclature: See separate page